Kakeya Upper Bounds for Neural Arbors

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Abstract

We establish a bridge between incidence geometry and neurobiological structure by showing that the classical joints problem provides a strict upper bound on the branching complexity of neural arbors. We refine this connection using bounded-degree incidence graphs, demonstrating that pruning, directionality, and conservation laws reduce the combinatorial complexity of intersections from $O(N^{3/2})$ to $O(N)$. In particular, dendritic trees modeled after Rall's framework naturally induce bounded-degree, acyclic incidence structures, leading to sharp linear upper bounds on branching complexity.

1 Introduction

The joints problem asks: given N lines in $\mathbb{R}^3$, how many points can occur where three non-coplanar lines intersect? The answer is:

$$|J| = O(N^{3/2}).$$

This bound reflects maximal combinatorial reuse of lines.

In contrast, neural arbors (e.g., dendritic trees) exhibit:

- directionality,
- acyclicity,
- conservation constraints.

We show that these constraints correspond to *bounded-degree incidence graphs*, which enforce linear scaling of branching complexity.

2 The Joints Problem

Definition 1. A *joint* is a point in $\mathbb{R}^3$ where at least three lines intersect with linearly independent directions.

Theorem 1 (Joints Theorem)**.** *For N lines in $\mathbb{R}^3$,*

$$|J| \leq CN^{3/2}.$$

This represents the unconstrained regime.

3 Incidence Graph Formulation

We encode the configuration as a bipartite graph.

Definition 2. The *incidence graph* $G = (L \cup J, E)$ is defined by:

- vertices: lines L and joints J,
- edges: $(\ell, x) \in E$ if line ℓ passes through joint x.

In the classical setting:

- degrees of line vertices can be large,
- reuse of lines across many joints is unrestricted.

This high-degree freedom drives the $N^{3/2}$ bound.

4 Neural Arbor Model

We now introduce constraints inspired by dendritic trees.

Definition 3. An *arbor* is a directed, acyclic graph embedded in $\mathbb{R}^3$ obtained from initial trajectories via:

- orientation (signal flow),
- pruning of backward segments,
- non-reuse of segments after branching.

4.1 Rall-Type Constraints

In addition, branching satisfies:

$$\sum_j d_j^{3/2} = \text{constant},$$

imposing physical limits on branching proliferation.

5 Bounded-Degree Incidence Graphs

The key structural difference is degree control.

Definition 4. An incidence graph is *bounded-degree* if each line vertex has degree at most D, where D is independent of N.

Proposition 1. *The incidence graph of a neural arbor is bounded-degree.*

Proof. After pruning:

- each trajectory can only branch forward,
- each segment participates in at most one downstream branching event,
- reuse is forbidden.

Thus each original trajectory contributes a bounded number of incidences, implying a uniform degree bound D. $\square$

6 Edge Bound

Theorem 2 (Arbor Bound)**.** *Let T be an arbor derived from N initial trajectories. Then the number of edges satisfies*

$$E \leq kN$$

for some constant $k \geq 2$, and hence

$$|J_{arb}| \leq kN.$$

Proof. Intersections subdivide trajectories into multiple segments. For example, two intersecting trajectories may produce up to four local segments.

However, pruning enforces:

- directionality,
- acyclicity,
- non-reuse.

These constraints ensure that each initial trajectory yields at most a bounded number of retained segments. Hence $E \leq kN$ for some constant $k \geq 2$.

Since branching points are bounded by the number of edges,

$$|J_{\text{arb}}| \leq E \leq kN.$$

□

7 Comparison with Classical Regime

$$O(N) \ll O(N^{3/2})$$

Thus bounded-degree incidence graphs suppress combinatorial explosion.

8 Graph-Theoretic Interpretation

In general incidence graphs:

- average degree can grow with N,
- dense subgraphs enable many joints.

In bounded-degree graphs:

- total edges scale linearly,
- number of possible joints is constrained.

This aligns neural arbors with sparse graph regimes.

9 Biological Interpretation

Neural systems impose:

- metabolic constraints,
- signal propagation limits,
- structural stability.

These translate mathematically into:

- bounded degree,
- acyclicity,
- constrained subdivision.

Hence biological networks cannot realize extremal incidence configurations.

10 Main Conclusion

Theorem 3 (Kakeya Upper Bound for Neural Arbors)**.** *Any neural arbor derived from N spatial trajectories satisfies*

$$|J_{arb}| \leq kN,$$

while the unconstrained upper bound satisfies

$$|J| \leq CN^{3/2}.$$

This establishes a strict gap between geometric possibility and biological realizability. Future work will exploit this gap to bound the complexity of the neural connectome.

Keywords

Kakeya problem, joints theorem, incidence geometry, bounded-degree graphs, neural arbors, dendritic trees, Rall model, connectome complexity.

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